

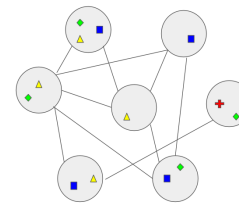


Research



Theory, software, and applications.

Partially observed Markov processes

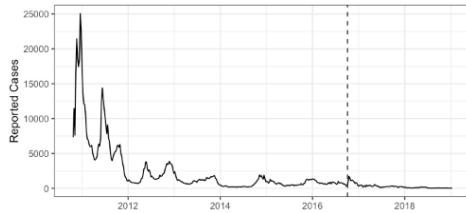
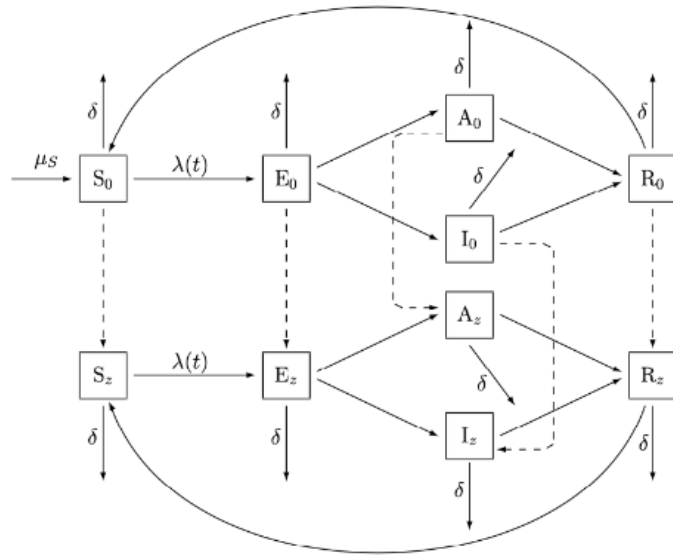


Models of dynamic systems:

Infectious disease dynamics.
Ecological population modeling.

Vaccination efforts in Haiti

Trout populations in Idaho



$$\mu_{\bullet Y_0}(t) = r_Y \bar{f}_Y(t) P_Y(t) \frac{d\Gamma(t)}{dt}$$

$$\mu_{\bullet I_0}(t) = \mu_{\bullet I_0}(t) = \left(r_I \bar{f}_I(t) P_{I,1}(t) + r_Y \bar{f}_Y(t) P_{I,2}(t) \right) \frac{d\Gamma(t)}{dt}$$

$$\mu_{\bullet S}(t) = M(t)$$

$$\mu_{Y_0 Y_1}(t) = \mu_{Y_1 Y_2}(t) = \mu_{Y_2 Y_3}(t) = 1 \text{ yr}$$

$$\mu_{I_0 I_1}(t) = \mu_{I_1 I_2}(t) = \mu_{I_2 I_3}(t) = 1 \text{ yr}$$

$$\mu_{Y_0 \bullet}(t) = \delta_Y + C_Y(t)$$

$$\mu_{I_0 \bullet}(t) = \delta_I + C_I(t)$$

$$\mu_{Y_1 \bullet}(t) = \mu_{Y_2 \bullet}(t) = \mu_{Y_3 \bullet}(t) = \delta_Y$$

$$\mu_{I_1 \bullet}(t) = \mu_{I_2 \bullet}(t) = \mu_{I_3 \bullet}(t) = \delta_I$$

$$\mu_{S \bullet}(t) = \delta_S \left[1 + c_h \left(S + \sum_{j=1}^3 (I_j(t) + Y_j(t)) \right) / K \right]$$